MIL-R-38327B(USAF)
23 February 1966
SUPERSEDING
MIL-R-38327A(USAF)
20 April 1965

MILITARY SPECIFICATION

RECEIVING SET, LORAN AN/ARN-78

1. SCOPE

1.1 This specification covers the requirements of one type of receiving set, designated as Receiving Set, LORAN AN/ARN-78.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; And Mounts, Vibration (For Use With Electronic Equipment in Aircraft)
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-T-5422	Testing, Environmental Aircraft Electronic Equipment
MIL-F-6106	Relays, Electric, Aircraft, General Specification For
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-C-8384	Connectors, Plug and Receptacle, Electrical (Molded Body) and Accessories, General Specification For
MIL-F-15160	Fuses; Instrument, Power, and Telephone
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification For
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification For
MIL-H-27894	Engineering Requirements for Aerospace Systems and Equipment

FSC 5826

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STANDARDS

Federal

FED-STD-5	Standard Guides for Preparation of Item Identification by Government Suppliers

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-415	Test Points and Test Facilities Design, Standard For
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-831	Test Reports, Preparation Of
MS3112	Connectors, Receptacle, Electric, Solder Type, Box Mounting Flange, No. 4 Holes, Bayonet Coupling
MS3126	Connectors, Plug, Electric, Crimp Type, Straight, Bayonet Coupling
MS25245	Connections, Input Power, Airborne Electronic Equipment, 28-Volt DC, 115-Volt Single-Phase AC, or 115/200 Volt 3-Phase AC, or Combinations
MS35174	Connector, Receptacle, Electrical, Series BNC, UG-291C/U
MS91405	Bases, Mounting, Large Size (For Use with Electronic Equipment in Aircraft)

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1. Preproduction. This specification makes provisions for preproduction testing.

3.2 Units. One complete Receiving Set, LORAN AN/ARN-78 shall consist of:

Item Nr	Quantity	Description	See Requirements
1	1 ea	Coupler, Antenna CU-1390/ARN-78	3.6.1
2	1 ea	Control, Receiver C-6430A/ARN-78	3.6.2
3	1 ea	Control, Indicator C-6431/ARN-78	3.6.3

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Item Nr	Quantity	Description	See Requirements
4	1 ea	Receiver, LORAN R-1298A/ARN -78	3.6.4
	1 ea	Mounting MT-3479/ARN-78	3.6.4.16
*5	1 ea	Indicator, Light-Time ID-1249/ARN-78	3.6.5

*This item is an optional unit.

3.3 General specification. The requirements of MIL-E-5400 shall apply as requirements of this specification, with additions or exceptions to the paragraphs of MIL-E-5400, as identified by similarity of paragraph headings, called out herein. When the two documents conflict, this specification shall govern.

3.3.1 Classification. The Receiving Set, LORAN AN/ARN-78 shall be class 1 as defined in MIL-E-5400.

3.3.2 Materials, parts, and processes

3.3.2.1 Definitions

3.3.2.1.1 Approval of nonstandard parts and materials. Nonstandard parts and materials approval shall be based upon the requirements specified in MIL-E-5400 for a category III equipment contract and the requirements specified herein.

3.3.2.1.1.1 Unless specifically requested by the procuring activity, submittal of approval requests for the use of nonstandard parts and materials, other than tubes and semiconductors, shall not be required. However, the contractor shall test all his own developed nonstandard parts and maintain a file of test data and engineering justification for the application of such parts. The testing of nonstandard parts by the contractor shall consist of those tests that he normally performs to assure that the parts are reliable and capable of performing the functions in which he plans to use them. The contractor shall request test data on nonstandard parts obtained from other sources and maintain a file of this data for a 1-year period from the date of test. The file of test data shall be accessible to engineering personnel of the procuring activity for their review as deemed necessary by such personnel. The number of the contractor's test report shall appear under the manufacturer's part number on the part data sheet. (See sample sheet in MIL-E-5400.)

3.3.2.1.1.2 Approval requests for the use of nonstandard and nonpreferred tubes and semiconductors shall be submitted in accordance with the requirements of MIL-E-5400.

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3.3.2.1.2 Preliminary list of electronic parts. The preliminary parts list shall include all parts and materials affecting the electrical characteristics of the equipment and shall be used as a check list for determining compliance with parts, material, and process requirements of MIL-E-5400. In addition to the requirements of MIL-E-5400, the preliminary parts list shall include the following:

a. Prime contractor's name and part number

b. Accurate identifying information in accordance with MIL-STD-16 and FED-STD-5

c. Accurate description of the part or material to include the following:

(1) Number of the standard specification (MIL, EIA, etc.) under which the part or material was manufactured.

(2) Electrical and mechanical ratings.

(3) Special features of construction.

(4) Materials used (type, grade, composition, etc.).

(5) Type of insulation, dielectric, etc.

(6) Type of finish or covering.

(7) Details of spring materials, contacts, etc.

(8) Details of tropicalization process.

3.3.2.2 Console control panels. All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781, type I. The color of lighting, red or white, shall be specified by the procuring activity.

3.3.2.2.1 Control panel lighting and pilot lamps on major units other than those in accordance with MIL-C-6781 shall be provided with brilliance controls.

3.3.2.3 Finishes

3.3.2.3.1 Cases and front panels. Cases and front panels of equipment installed in the cockpit area shall have a lusterless black finish.

3.3.2.4 Meters (electrical indicating instruments). Only ruggedized meters shall be acceptable.

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3.3.2.5 Tubes (electron) and solid state devices. The use of electron tubes shall be limited to display readout only.

3.3.3 Design and construction

3.3.3.1 Cases and mounting bases. Cases and mountings shall be in accordance with MIL-C-172. Other equipment unit form factors and the use of any type of rear-mounted electrical connector shall be subject to approval by the procuring activity. Class C items in accordance with MIL-C-172 shall be defined as nonstandard and shall be subject to the nonstandard parts approval procedures as specified herein.

3.3.3.2 Interference. Electronic interference shall be controlled in the basic design of all units of the Receiving Set, LORAN AN/ARN-78. The equipment shall also be designed to minimize susceptibility to interference from external sources.

3.3.3.3 Reliability, longevity, and maintainability

3.3.3.3.1 Reliability and longevity. Reliability and longevity of the Receiving Set, LORAN AN/ARN-78 shall be determined in accordance with MIL-R-26667.

3.3.3.3.1.1 MTBF. The mean-time-between failures (MTBF) of the receiving set shall not be less than 150 hours.

3.3.3.3.1.2 Longevity. The longevity of the receiving set shall not be less than 2000 hours.

3.3.3.3.2 Maintainability design requirements. General and specific maintainability requirements, maintainability program plan, data submission requirements, maintainability policy and general procedures for this program shall be in accordance with MIL-M-26512 and appendix A.

3.3.3.3.2.1 Quantitative. The mean active maintenance downtime at organizational and field levels shall not exceed 2 hours.

3.3.3.3.2.2 Maintenance analysis. At completion of the final paper design of the contract article, the contractor by design analyses, shall determine and establish the predicted mean active maintenance downtime at organizational and field levels of the equipment to be delivered. These values shall reflect compliance with the specified established quantitative requirement. The design analyses shall be in accordance with the procedures set forth in MIL-M-26512, appendix A. The predicted quantitative values shall be determined under the following conditions and assumptions:

a. Active maintenance downtime includes only that portion of downtime during which one or more technicians are actively engaged in system or equipment

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repair. Excluded is downtime due to maintenance scheduling and delays caused by deficient logistic support.

b. Part failure rate data that will be used in connection with the reliability predictions requirements and the estimated active maintenance downtime for each corrective and preventive maintenance task shall be used in the analyses.

c. The quantitative requirements are based on the time required by technicians to recognize, remove, diagnose, locate, repair, replace and checkout the system or equipment at organizational and field levels.

d. Facilities and environmental conditions will be typical of Air Force maintenance installations at organizational and field levels.

e. Active maintenance tasks will be performed by senior repairmen (skill level 5).

3.3.3.4 Maintenance provisions

3.3.3.4.1 Operational checkout provisions. When required by contract, special or unique items of test equipment shall be developed in accordance with MIL-T-21200.

3.3.3.4.2 Chassis test points. Built-in test facilities with no less than 250 test points shall be provided in Receiver, LORAN R-1298A/ARN-78 in accordance with MIL-STD-415. The test points selected shall provide capability with an external piece of test equipment for fault location purposes.

3.3.3.4.3 The equipment, in or out of the case, shall be so constructed that no damage to any unit or any undue distortion to any structural member will be caused by placing any flat side of the equipment on a flat surface.

3.3.3.5 Orientation. Unless otherwise specified, units that require mountings with vibration isolator shall be designed for normal installation with the mountings below the units. Nonisolated units designed for mounting directly to the aircraft structure shall permit installation in any orientation.

3.3.3.6 Overload protection. Fuses shall be style F05 in accordance with MIL-F-15160.

3.3.3.7 Pressurization. If not explosion-proof with the case open, pressurized equipment shall be protected by a safety switch to cut off

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all power in case excessive leakage permits the breathing of external air. This shall be provided whether the pressurization is applied on the ground prior to flight or is supplied by a pump installation which may be unable to keep up with the leakage rate.

3.3.3.8 Service conditions (electrical). The AN/ARN-78 receiving set shall be designated as a category B utilization equipment in accordance with MIL-STD-704, with exceptions or additions as specified herein.

3.3.3.8.1 A-c power source. The equipment shall perform as specified herein from a 108V to 121V, line-to-neutral, single or three-phase, 380 to 420 CPS, a-c power source. The choice of either single or three-phase input shall be at the discretion of the contractor; the equipment shall not be required to operate from both single and three-phase sources.

3.3.3.8.1.1 Continuous operation at 60 volts for 1 minute shall not result in equipment damage or malfunction.

3.3.3.8.2 D-c power source. Unless otherwise specified by the procuring activity as outlined in 3.3.3.8.2.1, d-c power shall be permitted only for such uses as relays, panel illumination and heaters.

3.3.3.8.2.1 The procuring activity may specify equipment that shall perform as specified herein when operated from 28-volt d-c power source only. The power required by the equipment shall not exceed 400 watts. Performance under emergency conditions of MIL-STD-704 shall not be required; however, equipment subjected to these emergency voltage conditions shall not be damaged.

3.3.3.8.3 Power ratings. The power required by the equipment shall not exceed the following:

a. A-c power:

(1) 525 va from a nominal 115V, 400 cps source at -55°C ambient.

(2) 325 va from a nominal 115V, 400 cps source at +55°C ambient.

b. D-c power:

(1) 30 watts maximum from a nominal 28V source after 3 minutes from the time power is applied to the equipment.

(2) 60 watts maximum during first 3 minutes after power is applied to the equipment.

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3.3.3.8.4 Power source transients. The equipment shall not be damaged or its performance degraded as a result of the transient test specified herein. Transients below 54 volts shall cause no damage though momentary interruptions of the tracking process may occur and reinitiate the search procedure.

3.3.3.8.5 Input power connections. Input power connections shall be in accordance with MS25245 or MIL-C-8384. The ON-OFF switch shall be located on the Control, Receiver C-6430A/ARN-78. A power relay meeting the requirements of MIL-R-6106 shall be included as required in the Receiver, LORAN R-1298A/ARN-78.

3.3.3.8.6 Warmup time. The maximum warmup time for the equipment shall not exceed 15 minutes under standard conditions and 20 minutes at extreme service conditions.

3.3.3.9 Service conditions (environmental). Class 1 equipment requirements of MIL-E-5400 shall apply, except as specified herein.

3.3.3.9.1 Temperature. The equipment shall perform as specified herein with relative humidity uncontrolled at operating temperature.

3.3.3.9.2 Altitude. For equipment used only in pressurized crew compartments, the minimum pressure altitude shall be 8.9 inches of mercury (30,000 feet altitude) operating, and 3.4 inches of mercury (50,000 feet altitude) nonoperating.

3.3.3.9.3 Humidity. The equipment shall perform as specified herein at relative humidity varying from 5 to 100 percent, including condensation at any temperature up to 50°C (122°F).

3.3.3.9.4 Vibration. The equipment shall operate as specified when subjected to vibration as specified in figure 5, curve IV of MIL-E-5400.

3.3.3.9.5 Sand and dust. For items normally installed within enclosed aircraft spaces, sand and dust requirements shall apply as a design objective. The sand and dust test shall not be required unless specifically designated for critical units.

3.3.3.10 Stability. The equipment shall operate with specified performance for at least 50 hours, continuously or intermittently, without adjustment of any controls which are not accessible during flight.

3.4 Design and construction

a. When used with the LORAN C system, the AN/ARN-78 receiving set shall be capable of simultaneous automatic track on two LORAN station pairs

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(master and any two slave stations). Automatic search or rapid lock-on techniques shall be so incorporated that the equipment shall successively locate and track the master and each of two slave signals with an operator furnishing only a limited amount of preset information in initiating the search operation. The equipment shall be capable of utilizing LORAN C information transmitted in the form of a group of 8 pulses (9 pulses for the master) phase coded in accordance with codes specified herein. The circuitry required for phase coding and multipulsed triggering functions shall be so incorporated in a single chassis that if more sophisticated and complex codes are subsequently employed, modification of the equipment could be accomplished by chassis substitution.

b. The equipment shall also be capable of operating with the 16 pulses of the LORAN D system when modified by substitution of plug-in assemblies. This modification shall be capable of being accomplished at the field shop level. The LORAN D information is transmitted in the form of a group of 16 pulses phase-coded in accordance with codes specified herein.

3.4.1 Integrated digital micro-molecular circuitry shall be used to the most practicable extent; however, no more than five circuit types shall be used. Cooling fans or other rotating equipment shall be prohibited.

3.4.2 Plug-in assemblies. Easy disconnect plug-in assemblies shall be utilized for digital micro-molecular circuits. No more than five circuits shall be packaged on a single assembly. The use of easy disconnect electronic modules shall be adopted where practical for improvement in maintainability and performance.

3.4.3 Fail-safe design. The Receiving Set, LORAN AN/ARN-78 shall incorporate "fail-safe" design features. Malfunctionioning of the equipment shall in no way contribute to the destruction of that equipment or any part of the weapon system which would result in loss of life.

3.4.4 Human factors. All equipment that is the immediate concern of its operator or of maintenance personnel shall be so designed and engineered as to result in the best possible man-machine combinations. This shall be accomplished within the necessary space, weight, and unit size limitations. The human factors data as outlined in MIL-H-27894 shall apply except that reports shall not be required. The contractor's proposed plan shall be subject to approval by the procuring activity prior to fabrication of the equipment.

3.5. Performance. The Receiving Set, LORAN AN/ARN-78 shall furnish accurate and reliable navigation information. The performance of the equipment shall be such that it can be operated and accurate readings obtained by persons having only limited training in its use. The equipment shall operate on one frequency channel of 100 kc, the nominal of 90 to 110 kc. Appropriate

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means shall be taken to prevent degradation of performance resulting from interference and noise as defined herein.

3.5.1 Effects of noise and interference on measurement accuracy. For any combination of amplitude of input signals and interference conditions with which the equipment is herein specified to operate, the error introduced by effects of noise interference or measurement of time difference between appropriate ground wave pulses shall be less than those specified in table I for 95 percent of the observations.

TABLE I. LORAN C

BASELINE VELOCITY (knots)	S/N WEAKEST SIGNAL (db)	AMPL. UNBAL. (db)	INTERFERING FREQUENCY (kc)	RATIO OF WEAKEST SIGNAL TO INTERFERING SIGNAL (db)	TIME DIFFERENCE (Nanoseconds)
1500	0	40			200
0	0	40			200
0	-20	40			900
0	20	0	>85<118	-35	300
0	20	0	<85>118	-40	300

3.5.2 Acceleration. The acceleration characteristics of the receiving set shall be such that it does not lose synchronization under the following aircraft conditions:

a. A 1g turn for a signal-to-noise ratio of 0 db.

b. 30 seconds after completion of the above manuever, in the presence of the above specified noise the system error shall have reduced to less than 300 nanoseconds. The accuracy shall not be degraded by the accelerations due to normal roll, pitch, and yaw of the aircraft.

c. The system shall maintain synchronization during a 1/4g turn at a signal-to-noise ratio of 20 db.

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3.5.3 Signal source. The equipment shall operate as specified when fed by a vertically polarized antenna with an effective height of at least 1 meter and resulting, with coaxial cable if used, in a capacitive input to the antenna coupler of a value between 70 and 300 picofarad (PF).

3.6 Details of units

3.6.1 Coupler, Antenna CU-1390/ARN-78. The antenna coupler shall contain the necessary circuits to properly match the antenna to the receiver and automatic notch filters for cw interference. Its internal components shall be protected from environmental conditions associated with installation in remote areas of the aircraft. The antenna coupler shall provide for:

a. Tuning capability to accomodate antenna characteristics

b. Coaxial cable feed to the receiver with a length up to 100 feet

c. Compatibility with a capacitive antenna between 70 and 300 PF

d. Automatic notch filters for cw interference. The notch filters shall be automatically adjustable over the range of 70 to 130 kc and shall provide at least 30 db suppression of A1 or F1 signals within the band of 85 to 115 kc and at least 25 db suppression of A1 or F1 signals within the bands of 70 to 85 kc and 115 to 130 kc. The notch shall be symmetrical at the -3 db points within ±2 db from 85 to 115 kc.

e. Coaxial cable feed to a communications receiver with minimum attenuation in the 2.0 to 30 mc band.

3.6.1.1 Size and weight. The over-all dimensions of the antenna coupler shall be as shown on figure 1. The antenna coupler weight shall not exceed 4.0 pounds.

3.6.1.2 Controls. Except for maintenance adjustments there shall be no controls located on the antenna coupler.

3.6.1.3 Electrical connections. Connections to external circuits shall be as shown on figure 2.

3.6.1.4 Mounting. The antenna coupler mounting dimensions shall be as shown on figure 1.

3.6.2 Control, Receiver C-6430A/ARN-78. This control shall provide for operation of the LORAN receiver. The receiver control shall display the time difference readings in digital form to a resolution of 0.1 microsecond. The time difference indication shall be 00,000.0 during master search and shall indicate the search time difference during slave search.

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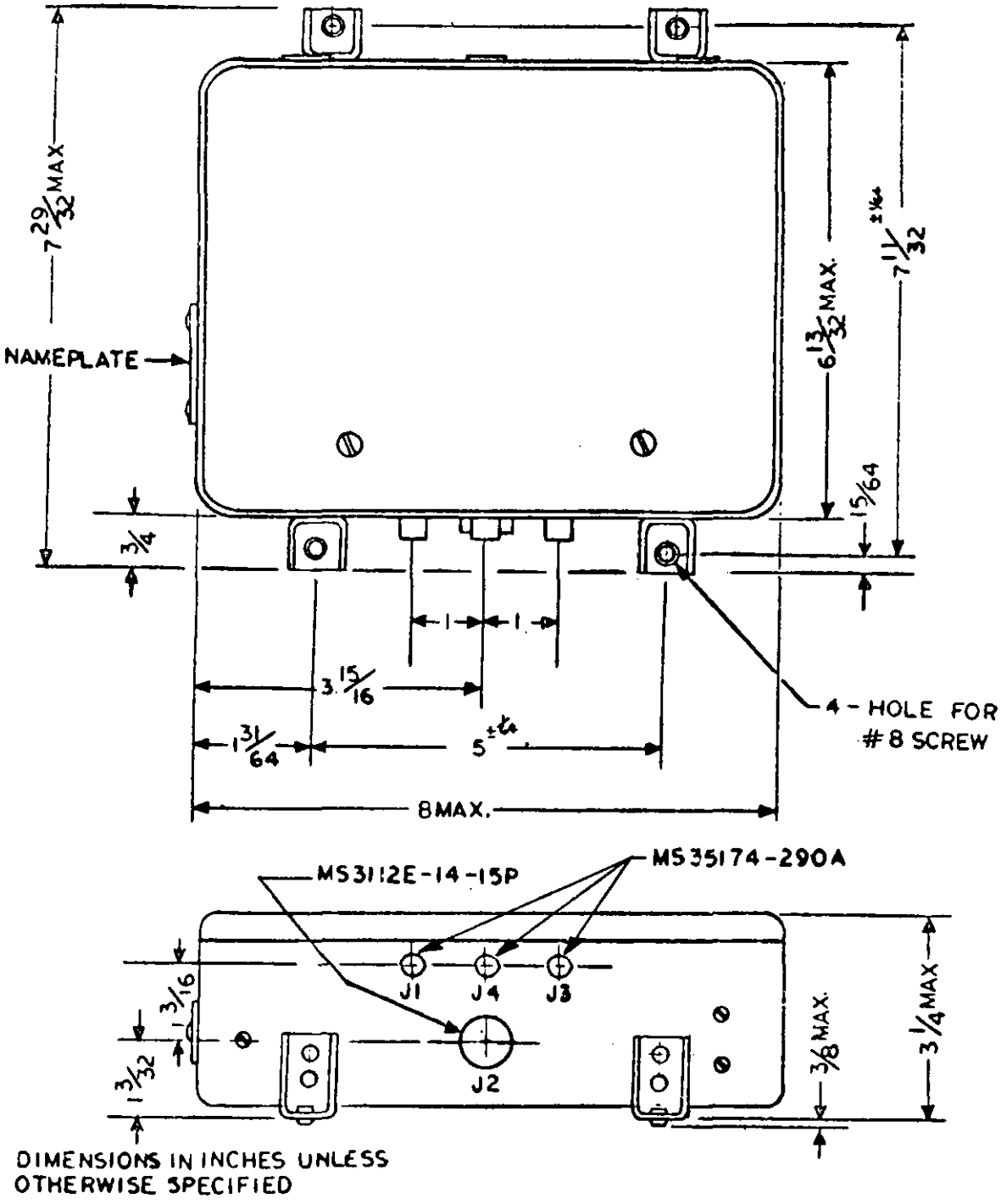


Figure 1. Coupler, Antenna CU-1390/ARN-78

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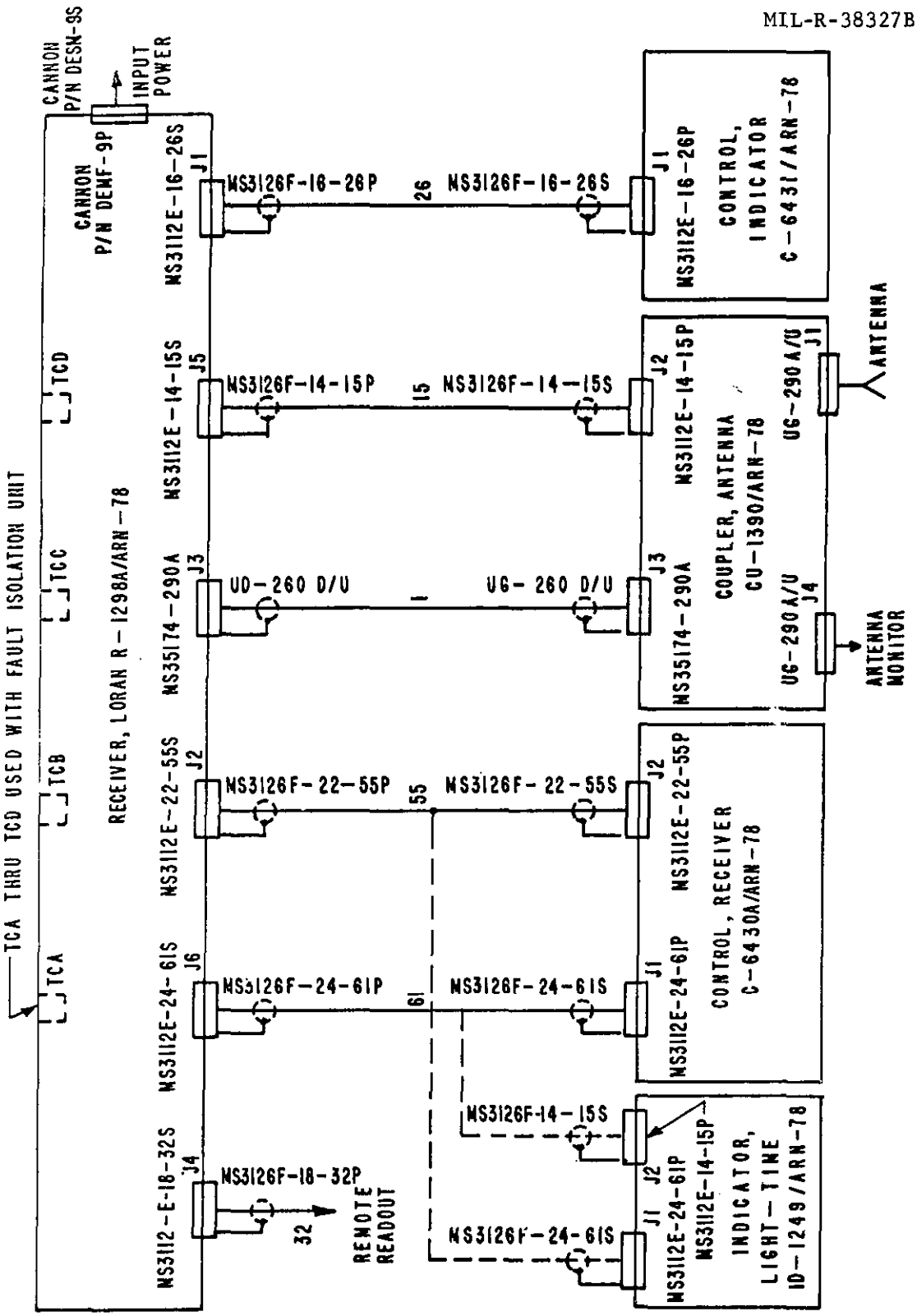


Figure 2. Block Cabling Diagram Receiving Set, Loran AN/ARN-78

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The display shall include a combined cycle and envelope reading without ambiguity or envelope cycle discrepancy. The following controls shall be available to the operator:

a. Power (Off-On)

b. Basic Rates (SS, SL, SH, S, and L)

(1) LORAN C: SS, SL, SH, S, and L. Uses 8 pulses per slave station and 9 pulses per master station.

(2) LORAN D: S, and L. Uses 16 pulses per master and slave stations. The LORAN D basic rates enable the AN/ARN-78 receiving set to operate in the LORAN D chain (using the alternate odd pulses as shown in the LORAN D phase coding chart, see 6.3.2.10.2).

c. Specific Rates (0, 1, 2, 3, 4, 5, 6, and 7)

d. Time Difference Preset A

e. Time Difference Preset B

f. Read-Reset

g. Display Brightness.

The display shall include "plus" (+) and "minus" (-) lamps that indicate 10-microsecond changes in the time difference reading due to envelope synchronization changes. The reset switch shall have a position that extinguishes these lights; the lights shall extinguish automatically as a "plus" jump is followed by a "minus" jump, or vice versa.

3.6.2.1 Size and weight. The over-all dimensions of the receiver control shall be as shown on figure 3. The receiver control weight shall not exceed 2.9 pounds.

3.6.2.2 Electrical connections. Connections to external circuits shall be as shown on figure 2.

3.6.2.3 Mounting. The receiver control mounting dimensions shall be as shown on figure 3.

3.6.3 Control, Indicator C-6431/ARN-78. This control shall enable extended LORAN C range operation by providing the following:

a. Ability to override automatic operation by operator to permit skywave operation including indication of synchronization on skywave (GRD AVAIL).

b. Forced tracking mode to permit optimized operation under extreme range conditions.

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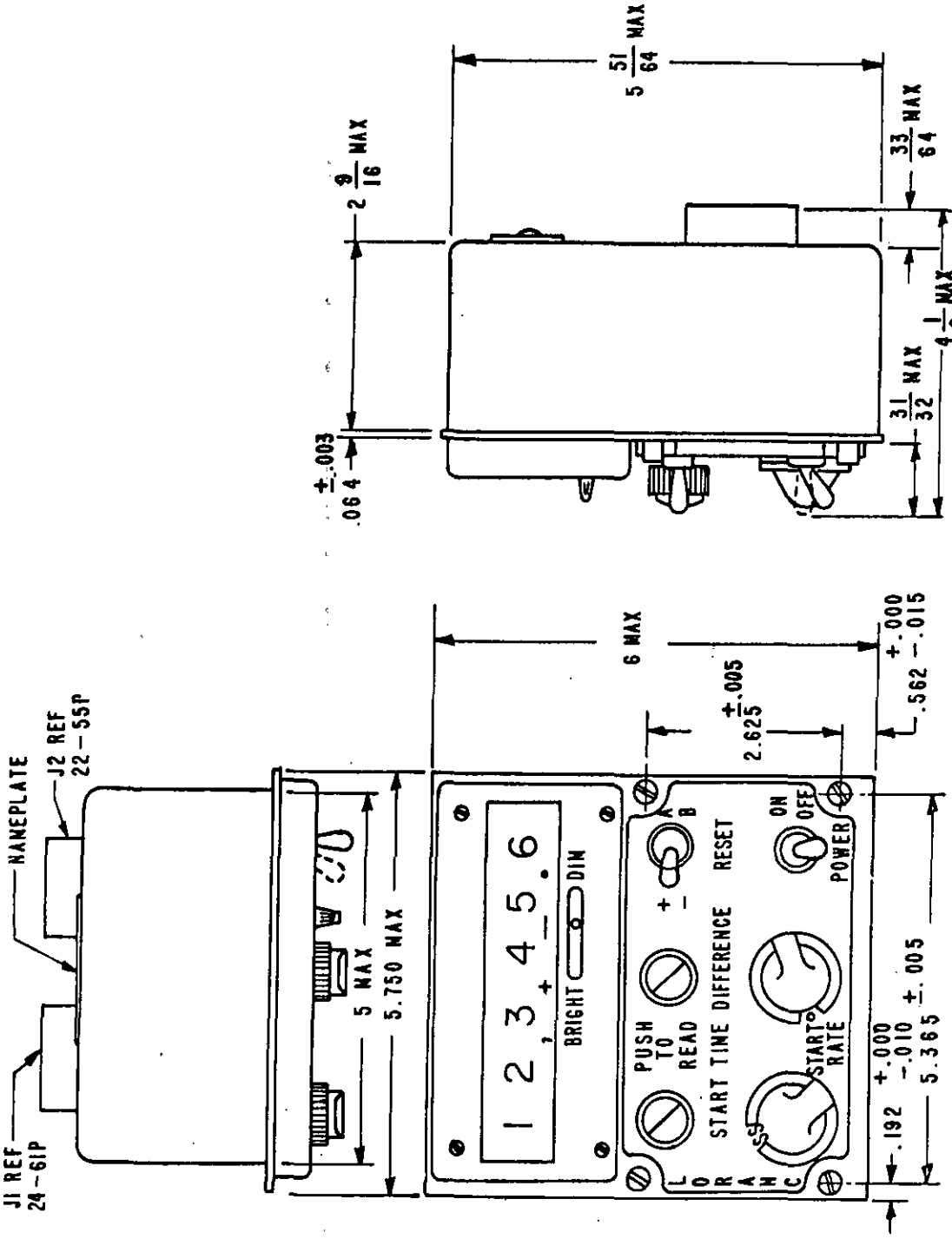


Figure 3. Control, Receiver C-6430A/ARN-78

DIMENSIONS IN INCHES

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c. Continuous display of either time difference.

d. Warning of incorrect transmitter operation (BLINK).

e. Warning indication of weak signal or excessive interference (RCVR OVLD).

3.6.3.1 Controls and displays. The indicator control shall provide the following controls and displays to an operator:

a. Controls:

(1) Mode selection (Auto. Manual)

(2) Time difference display selection (A, B or alternating A/B)

(3) Time difference manual slew

(4) Station selection for slew and indicators.

b. Displays:

(1) Receiver overload (RCVR OVLD)

(2) Ground available (GRD AVAIL)

(3) BLINK

3.6.3.2 Size and weight. The over-all dimensions of the indicator control shall be as shown on figure 4. The maximum weight of the indicator control shall not exceed 0.9 pounds.

3.6.3.3 Electrical connections. Connections to external circuits shall be as shown on figure 2.

3.6.3.4 Mounting. Mounting dimensions shall be as shown on figure 4.

3.6.4 Receiver, LORAN R-1298A/ARN-78

a. The receiver shall be designed to receive, detect, measure, and indicate time differences required for LORAN "C" navigation used in conjunction with the antenna coupler, indicator control, and receiver control.

b. The receiver shall contain the necessary input power connector and generate necessary power supply voltage for all units of the receiving set.

c. The receiver shall be so designed that operation on 16 LORAN D pulses may be obtained when the equipment is only modified by substitution of plug-in assemblies. In addition, the equipment shall be capable of being modified by substitution of plug-in assemblies to accept externally generated velocity information as specified herein.

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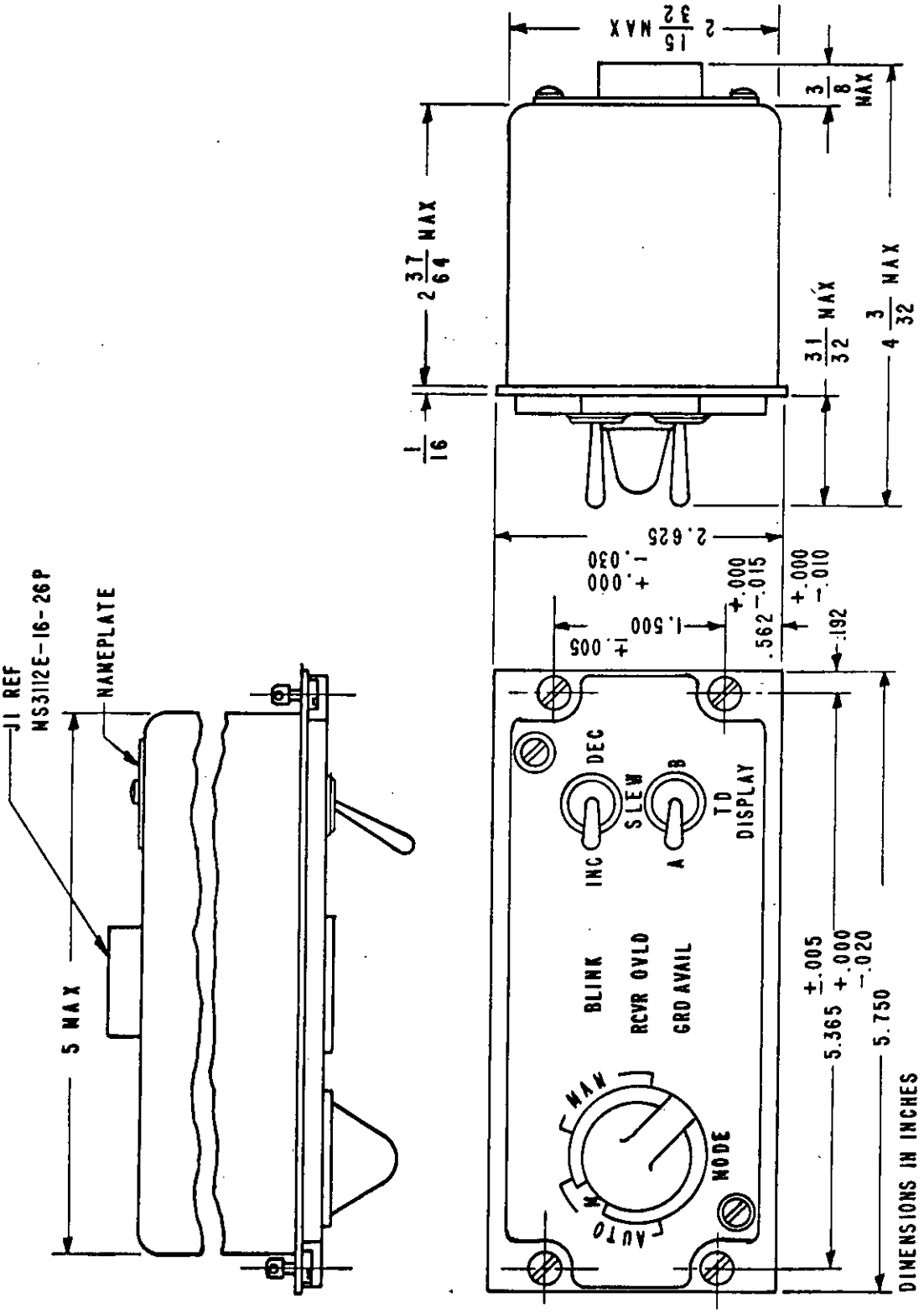


Figure 4. Control, Indicator C-6431/ARN-78

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d. The receiver shall have provision for pure binary readout of time difference to an external computer, by plug-in of suitable components.

3.6.4.1 Size and weight. The over-all dimensions of the LORAN receiver shall be as shown on figure 5. The LORAN receiver weight shall not exceed 26.5 pounds.

3.6.4.2 Electrical connections. External connections to the receiver shall be as shown on figure 2.

3.6.4.3 Receiver bandwidth. The bandwidth required for operation of LORAN C and LORAN D is defined by the following selectivity characteristics (fo = 100 kc):

a. Tracking Mode:

Attenuation (db)	LORAN C:	Bandwidth (kc)
-3		23 ±2
-40		145 ±45
	LORAN D:	
-3		16 ±2
-40		115 ±30

b. Search or Pre-Lock-on-Mode, LORAN C and LORAN D:

Attenuation (db)	Bandwidth (kc)
-3	5 ±1
-20	25 ±8

3.6.4.4 Overload and recovery time. The receiver shall be designed to have as fast a recovery time after receiver saturation as is practicable. In any event it shall recover in not more than 50 microseconds after a signal causing saturation is removed.

3.6.4.5 Sensitivity. The receiver shall operate properly in the tracking mode of 5 microvolt signals fed through a 100 pf dummy antenna to the input of the antenna coupler.

3.6.4.6 Maximum signal input voltages. The receiver shall be capable of operating properly with a maximum signal of 5 volts through the dummy antenna specified in 3.6.4.5 to the input of the antenna coupler.

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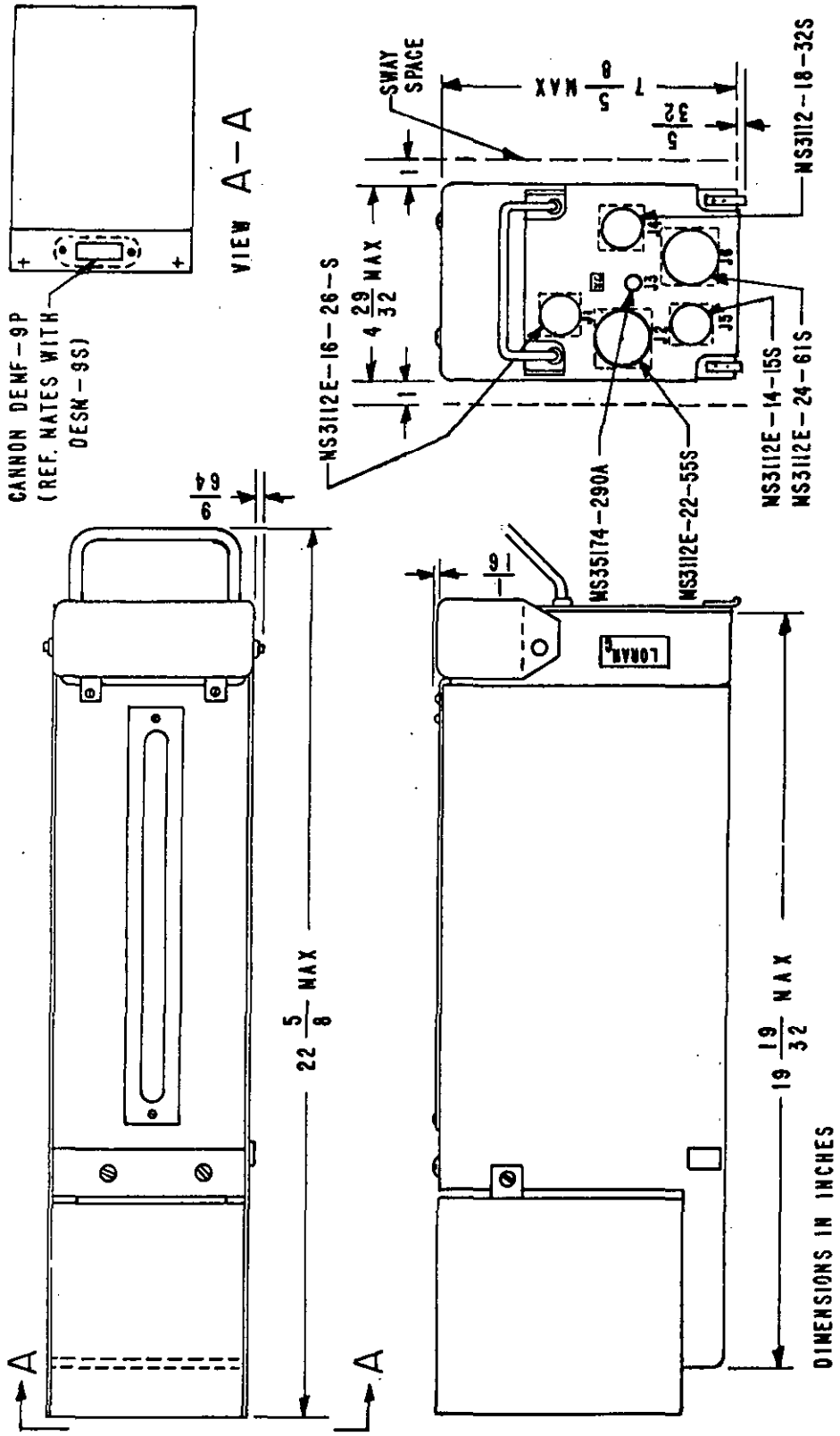


Figure 5. Receiver, Loran R-1298A/ARN-78

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3.6.4.7 Signal amplitude balance. Signal amplitude balance shall be automatically maintained over a range of amplitude unbalance of at least 10,000 to 1 (80 db).

3.6.4.8 Timer oscillator. Only one oscillator shall be used in the design of the LORAN C timing section of the receiver. The heater circuits for the oscillator shall be controlled separately to provide for standby operation.

3.6.4.8.1 Manual frequency adjustment. A manual frequency or drift control shall be provided, as a service adjustment, the use of which will enable compensatory adjustments for slight variations in the timer oscillator frequency. This control shall permit adjustment of the crystal oscillator over a range of ± 3 parts in 10^6

3.6.4.9 Frequency dividers. Frequency dividers utilized to derive the various LORAN repetition rates from the crystal oscillator shall be of a highly stable type. Timing jumps shall not occur when tested in accordance with all requirements specified herein except zero voltage condition in 115V a-c supply per MIL-STD-704. Timing waveforms over the entire pulse repetition interval shall be continuous. Discontinuity in the timing waveforms, as required to obtain specific pulse repetition intervals, shall be accomplished only at the end of the interval.

3.6.4.10 Time difference measurement. It shall be possible to measure the time difference between master and slave LORAN pulse envelopes when the delay between the last master and first slave pulse and between adjacent slaves is as low as 2500 microseconds. It shall be possible to measure the time difference between master and slave pulses when the delay between the last slave and the first master pulse is as small as 3000 microseconds. (This specifies an upper limit of slave delay.) It shall be possible to measure any value of delay between these limits.

3.6.4.10.1 Time difference resolution. Individual signal time of arrival measurements shall exhibit a resolution of 0.05 microsecond.

3.6.4.10.2 Over-all instrumental accuracy. Static error in time difference measurement read from the time difference display in the ranges specified shall not exceed 0.1 microsecond due to nonlinearity of time difference circuits or to a differential signal amplitude of 80 db.

3.6.4.11 Time sharing of repetition interval. The repetition interval will normally be shared by the transmissions from a master station and up to three slave stations. It is not possible to fix definite time or percentage lengths for the slave time intervals since these will be dictated by LORAN system type, deployment, repetition rate assignment and

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other factors. Since it must be possible for the slave servo systems to look on and track any of the slave station signals regardless of their location in the total time interval, extreme flexibility shall be designed into the equipment.

3.6.4.12 Synchronous detection. Detection of the r-f signal shall be of the synchronous, or coherent, type. The received LORAN signals shall be cross-correlated against a reference signal derived from the basic oscillator.

3.6.4.12.1 A circuit shall be provided to automatically sense synchronous cw interference and to automatically reduce the effects of such synchronous cw interference. This circuit shall protect the cycle, envelope, automatic jam control, and guard and blink circuits.

3.6.4.13 Signal strobing. The signal shall be strobed by the sampler to eliminate skywave interference.

3.6.4.13.1 The equipment shall be designed to accept a plug-in crossing rate blanking circuit to minimize the effect on accuracy of strong LORAN signals on rates other than that being tracked.

3.6.4.14 Search synchronization and clear guard

3.6.4.14.1 The equipment shall be capable of automatic signal synchronization and shall have a facility incorporated to ensure that synchronization is on the ground wave and not on a skywave. For purposes of demonstrating these functions a skywave to ground wave ratio shall be a minimum of 10 db with skywave delayed a minimum of 40 microseconds.

3.6.4.14.2 The equipment shall be capable of automatic synchronization under all signal amplitude conditions, signal-to-noise ratios, and velocities specified herein, except the signal-to-noise ratio shall be limited to -15 db at velocities above 1200 knots. Provisions shall be incorporated for initially inserting and storing the expected time difference thereby reducing lock-on time. Slave search shall automatically scan over a time difference range not to exceed 6000 microseconds with the start of the search window being continuously selectable. Search shall not exceed 10 minutes and shall ordinarily be accomplished using the narrow band setting provided in the receiver. When the narrow band selectivity is used for search, means shall be incorporated that prevent loss of synchronization upon switching to the wide band selectivity. In addition, the equipment shall include provision for automatically returning to slave search if a slave signal is lost or automatically return to master and slave search if a master signal is lost. The equipment shall automatically return to search whenever a new LORAN rate is selected, and, by use of the reset switch, it shall be possible to selectively initiate search for a new slave station without loss of master track.

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3.6.4.15 Display reading ambiguity resolution. Cycle and envelope reading shall be displayed without ambiguity or envelope cycle discrepancy. The cycle and envelope ambiguity resolution shall not be dependent upon an absolute fixed time relationship between carrier phase and the pulse envelope.

3.6.4.16 Mounting MT-3479/ARN-78. A mounting designed in accordance with MIL-C-172 shall be provided for the R-1298A/ARN-78 receiver. Electrical connections to the mounting shall be made by fastening the input power connector to the rear of the mounting. The mounting shall use self-locking clamps.

3.6.4.16.1 Size and weight. The over-all dimensions of the receiver mounting shall be as shown on figure 6. The weight of the mounting shall not exceed 2.3 pounds.

3.6.5 Indicator, Light-Time ID-1249/ARN-78. The light-time indicator shall contain circuitry for a 6-digit decimal readout and time difference display selection. The following controls shall be available to the operator:

a. Mechanical dimmer for 6-digit decimal readout

b. T/D display selector (A, A/B, B).

The slave indicator shall reproduce the time difference display of the receiver control but shall permit a reduction to a nonalternating display. The time differences shall be displayed as a number using decimal indicators with the smallest digit equal to 0.1 microsecond and shall be selected by the T/D selector switch.

3.6.5.1 Size and weight. The over-all dimensions of the light-time indicator shall be as shown on figure 7. The weight of the indicator shall not exceed 1.8 pounds.

3.6.5.2 Electrical connections. External connections to the receiver shall be as shown on figure 2.

3.7 Total weight. The total weight of the Receiving Set, LORAN AN/ARN-78, excluding cables, and shock mounts, shall be a minimum consistent with good design and shall not exceed 36.1 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to

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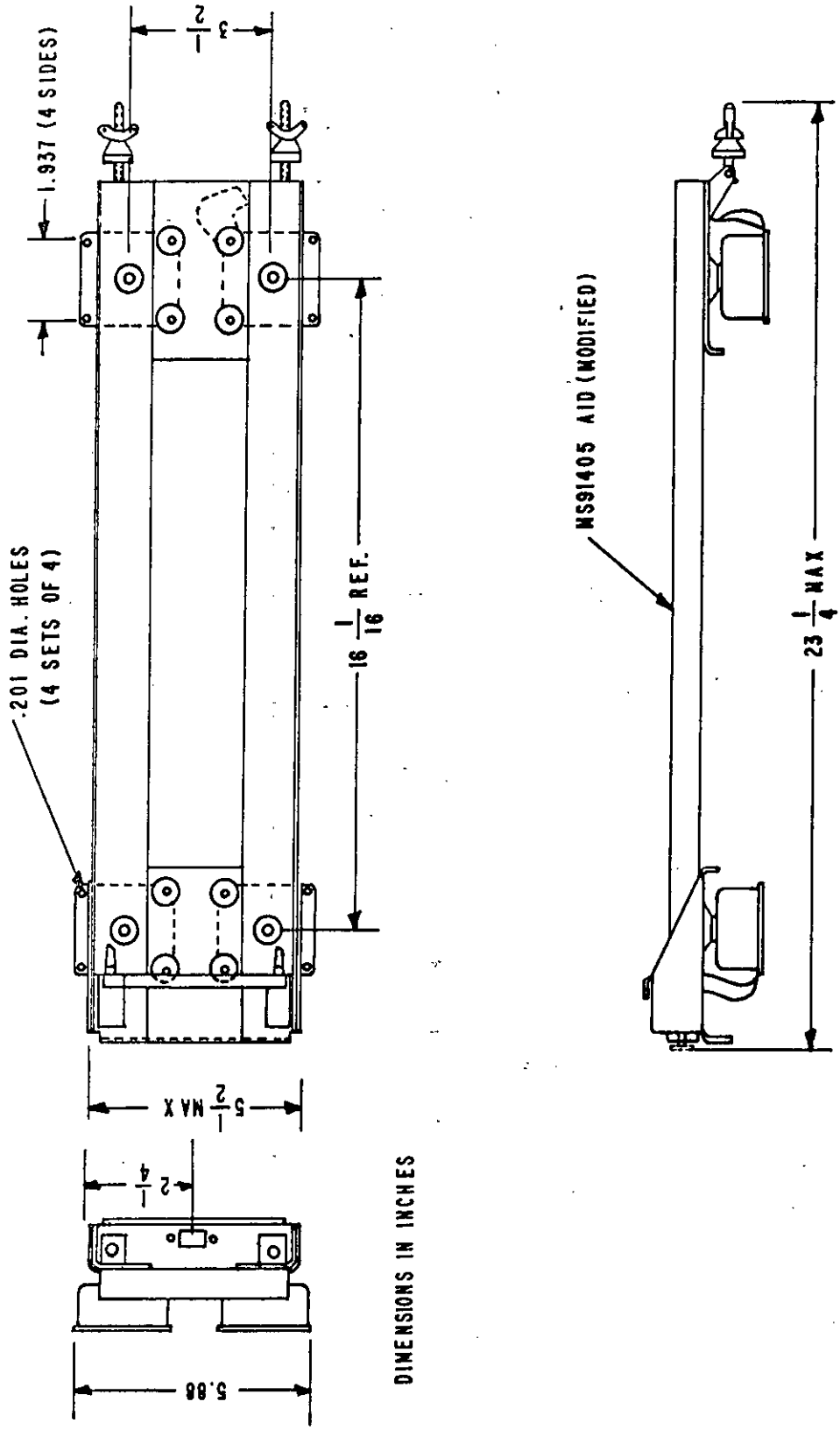


Figure 6. Mounting MT-3479/ARN-78

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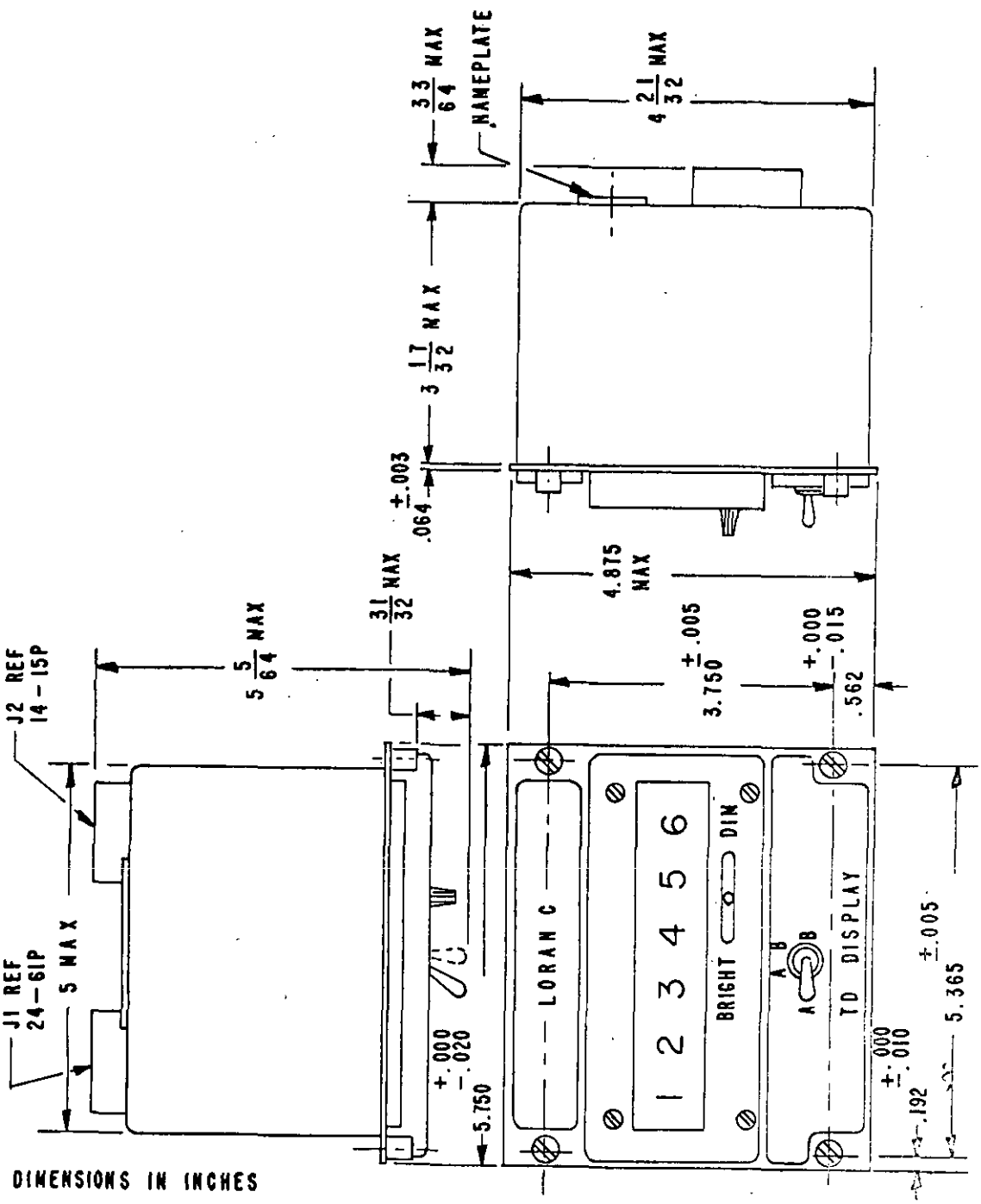


Figure 7. Indicator, Light-Time ID-1249/ARN-78

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perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The quality assurance provisions of MIL-E-5400 and the requirements specified herein shall be applicable to this specification. When the two documents conflict, this specification shall govern.

4.2.1 Classification of tests. The inspection and testing of AN/ARN-78 receiving set shall be classified as follows:

a. Preproduction tests

b. Acceptance tests

4.3 Test conditions. Unless otherwise specified, the equipment shall be subjected to the tests under the following test conditions:

a.	Temperature:	Room ambient (25°C, 15°C or -5°C)
b.	Altitude:	Normal ground
c.	Vibration:	None
d.	Humidity:	Room ambient
e.	Power:	115V, 400 cps ac; 28V dc.

4.4 Preproduction testing

4.4.1 Preproduction test samples. The preproduction tests samples shall consist of models representative of the production equipment. They shall be tested under the test conditions specified herein and at the point designated by the contract.

4.4.1.1 Sampling instructions. The contractor shall conduct tests on 11 preproduction sample equipments to determine that the design of the equipments meet the requirements specified herein. The contractor's test shall be conducted in accordance with the test procedures specified herein.

4.4.1.1.1 Testing schedule. The 11 sample equipments shall be processed as follows:

a. Sample Nr 1. Sample nr 1 shall be subjected to the following tests:

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Tests	Applicable Paragraph
Examination of product	4.6.1
Operational tests	4.6.2
Environmental tests as follows:	4.6.3
Shock	4.6.3.4
Explosion	4.6.3.6

b. Sample Nr 2. Sample nr 2 shall be subjected to the following tests:

Tests	Applicable Paragraph
Examination of product	4.6.1
Electrical tests	4.6.2.1
Environmental tests as follows:	
Temperature-altitude	4.6.3.3
Humidity	4.6.3.5
Vibration	4.6.3.4
Equipment stability	4.6.4

c. Sample Nr 3. Sample nr 3 shall be subjected to the following tests:

Tests	Applicable Paragraph
Examination of product	4.6.1
Electrical tests	4.6.2.1
Power tests	4.6.2.2
Power source transient	4.6.2.4

d. Seven equipments shall be subjected to the reliability demonstration test specified in 4.6.5. Two of the seven equipments shall be continually tested for longevity as specified in 4.6.6.

e. One equipment shall be subjected to the maintainability demonstration test specified in 4.6.7. The equipment shall be subjected to examination of product, 4.6.1, and electrical tests, 4.6.2.1, prior to, and after, the maintainability demonstration test.

4.4.2 Preproduction tests. Preproduction tests shall consist of the tests specified under 4.6, subjected to the test conditions specified herein.

4.5 Acceptance tests. Acceptance tests shall consist of individual tests and sampling plans and tests.

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4.5.1 Individual tests. Each Receiving Set, LORAN AN/ARN-78 shall be subjected to the following tests:

Test	Applicable Paragraph
Examination of product	4.6.1
Electrical tests	4.6.2.1
Power tests	4.6.2.2

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Production reliability acceptance tests shall be conducted in accordance with the requirements of test level III, table 2.3 of MIL-R-26667. The test program shall be in accordance with table II.

Table II. Production Reliability Test Program

Quantity ordered per year per contractor	Number of Reliability test programs per increment*	Total test programs per No. ordered per year
1 to 100	1	1
101 to 2002 2	1	2
Over 200	One from each additional 100 or fraction thereof on order	Variable

*A reliability test program shall consist of a quantity of at least 4 and not more than 7 units to be tested. The actual number shall be that which is most economical.

4.5.2.1.1 Accept-reject criteria. Accept-reject criteria shall be in accordance with table 5.2 of MIL-R-26667.

4.5.2.2 Sampling plan B. Quantities of Receiving Set, LORAN AN/ARN-78 shall be selected at random in accordance with table III and subjected to the sampling tests of 4.5.2.2.1.

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Table III. Sampling Plan B

Quantity of Consecutively Produced items on order by Groups Per Contractor	Number tested per Increment	Total tested per number ordered per year
1 to 25	1	1
26 to 50	1	2
51 to 75	1	3
76 to 125	1	4
126 to 200	1	5
201 to 300	1	6
Over 300	One from each additional 100 or fraction thereof on order	Variable

4.5.2.2.1 Sampling tests. Sampling tests shall consist of the following:

Tests	Applicable Paragraph
Environmental tests as follows:	4.6.3
Temperature-altitude	4.6.3.3
Vibration	4.6.3.4
Humidity	4.6.3.5

4.5.2.2.2 Rejection and retest. When one equipment from a production run fails to meet the specification, no equipments still on hand or produced later shall be accepted until the extent and cause of failure are determined and corrected.

4.5.2.2.3 Continuation of tests. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of equipments on hand or produced later shall not be made until it is determined that equipments meet all the requirements of the specification.

4.5.3 Defects in equipments already accepted. The investigation of a test failure could indicate that defects may exist in equipments already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 Test methods

4.6.1 Examination of product. The Receiving Set, LORAN AN/ARN-78 shall be visually inspected to verify that the materials, design and construction, necessary mechanical measurements, marking, and workmanship comply with the requirements of this specification.

4.6.2 Operational tests

4.6.2.1 Electrical tests. The procedures and application of electrical tests to determine proper performance and compliance with the requirements of this specification shall be prepared by the contractor and subject to approval by the procuring activity. The electrical tests shall be performed prior to the other specification and is ready to proceed with the other tests. Unless otherwise specified, the tests shall be conducted on the complete equipment.

4.6.2.2 Power tests. The Receiving Set, LORAN AN/ARN-78 shall be subjected to tests that will demonstrate compliance with the power supply requirements of this specification. These tests shall include tests as outlined in 4.6.2.1, with the power sources adjusted to extreme limits of specified voltages and frequency.

4.6.2.2.1 Primary power. Compliance with power ratings shall be ascertained concurrently with the power factor test of 4.6.2.2.1.1.

4.6.2.2.1.1 Power factor test. Compliance with the equipment power factor requirements of MIL-STD-704 shall be ascertained using the following instruments, or approved equal:

Wattmeter:	Weston model 310, form 3
Voltmeter:	Weston model 341
Ammeter:	Weston model 370

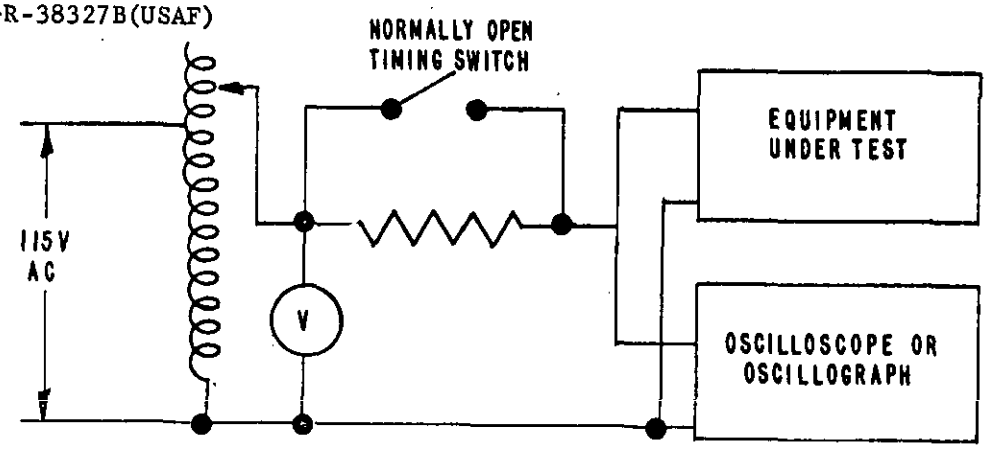
The sign of the power factor shall be determined by connecting suitable capacitors across the power input while the equipment is in operation, and measuring its effect on the resultant power factor.

4.6.2.3 Interference. Compliance with electrical interference and susceptibility requirements shall be checked in accordance with MIL-I-6181.

4.6.2.4 Power source transient response. The equipment shall be tested for transient response in accordance with figures 8 and 9.

4.6.3 Environmental tests. Environmental testing shall be in accordance with class 1 requirements of MIL-T-5422 with the following additions or exceptions. When the two documents conflict, this specification shall govern.

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The power source rating in kva should be considerably greater than the load kva in order to avoid difficulty in the determination of the effective voltage during the transient. Each of the four transients shall be repeated at least three times. The interval between transients shall not be less than 30 seconds. Tolerances shall be ± 2% on voltages and ± 10% on time intervals.

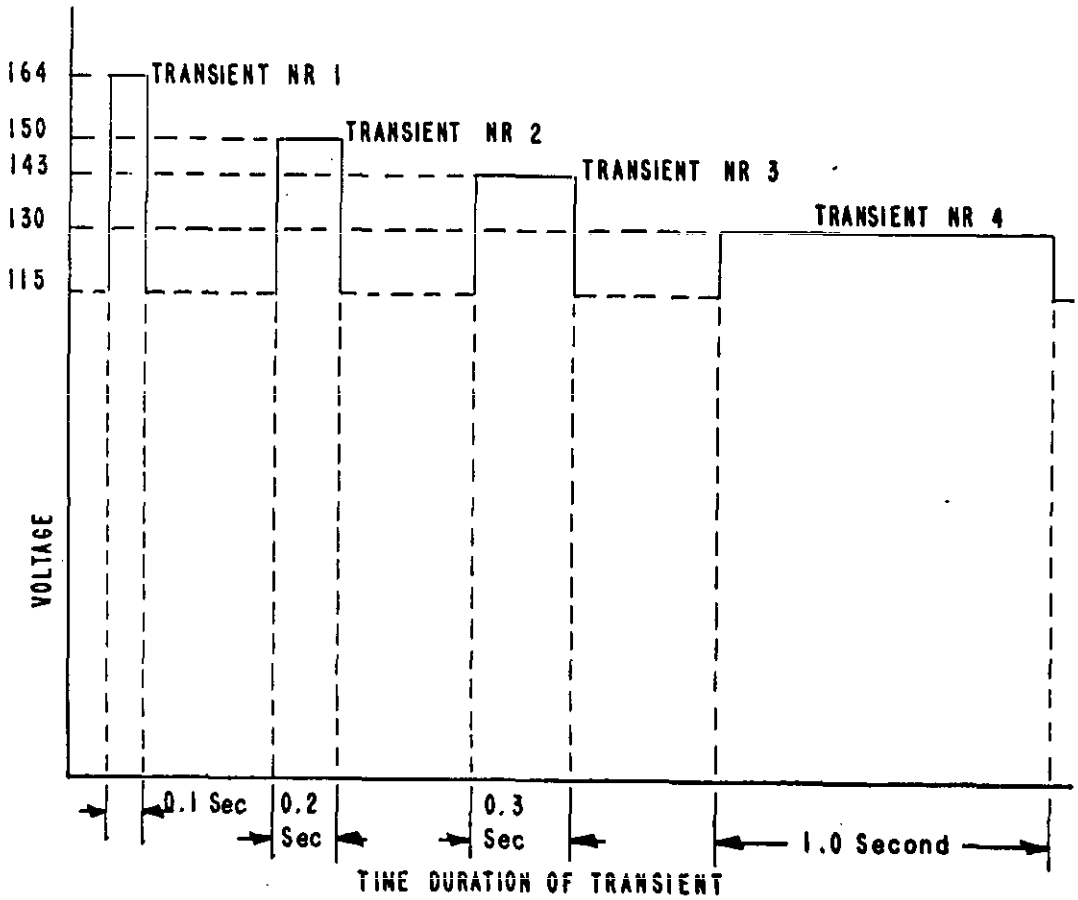
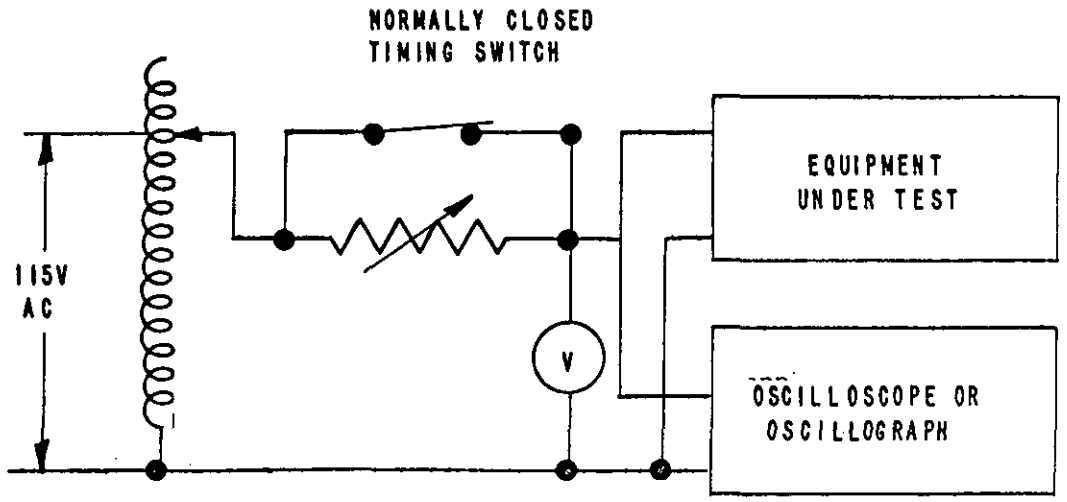


Figure 8. Overvoltage Transient Tests

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The requirements specified on figure 8 shall also apply to tests with these low voltage transients.

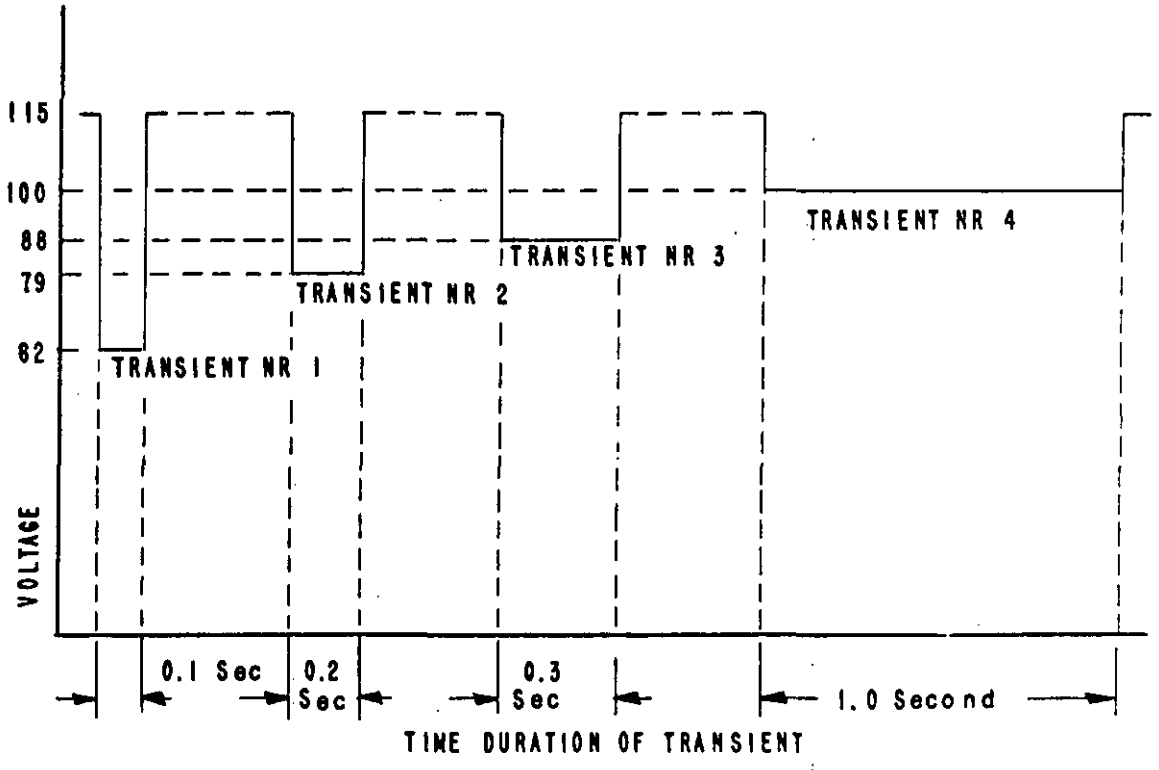


Figure 9. Undervoltage Transient Tests

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4.6.3.1 Salt spray, sand and dust, and fungus resistance tests shall not apply unless specifically designated for critical units. The explosion test shall not be required on pressure-sealed units.

4.6.3.2 Supplemental procedures. The contractor's test procedures, when approved, shall form a part of this specification.

4.6.3.3 Temperature and altitude tests. Units installed only in pressurized crew compartments shall be tested to a minimum pressure of 8.9 inches of mercury (30,000 feet altitude) for required performance, and nonoperating to 3.4 inches of mercury (50,000 feet altitude).

4.6.3.4 Vibration tests

4.6.3.4.1 Vibration isolated equipment with isolators removed. When equipment is vibrated with external isolators removed, it shall be nonoperating during the 10-minute periods at each resonant frequency.

4.6.3.5 Humidity test

4.6.3.5.1 Test installation. The equipment shall be placed in a vented test chamber in a manner simulating installed conditions whenever possible. The equipment shall be placed on an inert, interposing material such as loosely woven glass cloth to prevent the development of corrosion due to contact of dissimilar metals. The equipment shall be protected against dripping of condensed moisture by an antidrip device that will not impair the flow of air through the test compartment. An example of this antidrip device is, though not limited to, a gabled or sloped roof.

4.6.3.5.2 Test procedure. Commercial distilled water or tap water shall be used to obtain the necessary humidity. When wet and dry bulb thermometers are used to determine relative humidity, the minimum air velocity over the wet bulb shall be 900 feet per minute. The temperature within the test chamber shall be raised from prevailing room temperature, 20 C to 38 C (68 F to 100.4 F) to 50 C (122 F) and shall be maintained for 6 successive hours. At the conclusion of this 6-hour period, the temperature shall be returned to prevailing room temperature, at as uniform a rate as is possible, over a 16- to 17-hour period. This constitutes one temperature cycle. The relative humidity shall be 95±5 percent throughout each of the 24-hour temperature cycles, except that a transitory minimum relative humidity of 85 percent shall be permitted during the down cycle of temperature. The temperature cycle shall be repeated 10 consecutive times.

4.6.3.5.3 Comparison test. At the conclusion of the low stage of each temperature cycle and after removal of excess moisture by inverting or tilting the equipment and wiping it with a soft, clean cloth, the equipment shall be operated for a period not to exceed 1 hour and checked

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against the initial performance record. Unless the preliminary checks indicate that the equipment will be endangered by further testing, the equipment shall be subjected to the electrical tests specified in 4.6.2.1. If the equipment is found to be on the verge of failure and if testing will endanger the equipment, it shall be dried in a drying oven capable of maintaining a temperature of 37.7 C (100 F) for a maximum period of 4 hours. Check shall be made during this drying period to determine the recovery trends of the equipment. At the conclusion of the drying period, 4 hours of any cycle, the equipment shall be operated and checked against the initial performance record. If the equipment does not conform to specification tolerances, or is nonoperable, or excessively deteriorated, it shall be considered as having failed to pass the humidity tests.

4.6.3.6 Explosion test. Explosion tests shall be in accordance with MIL-T-5422, procedure I.

4.6.4 Equipment stability. All of the units required to make up a complete Receiving Set, LORAN AN/ARN-78 shall be put into operation with all dust covers and shock mountings in place. After the equipment has operated for an initial 30-minute warmup period, it shall be adjusted for optimum performance. The equipment shall be operated continuously for a period of 50 hours. Changes in performance that occur during the 50-hour test shall be noted in order to determine compliance with the stability requirements of Section 3.

4.6.5 Reliability demonstration test. The reliability demonstration test shall be conducted in accordance with test level III, table 2.3 of MIL-R-26667. Accept-reject criteria shall be in accordance with table 5.2 of MIL-R-26667.

4.6.6 Longevity test. An accumulable longevity test shall be conducted in accordance with the requirements of 4.6.5.

4.6.7 Maintainability. Maintainability tests and demonstration shall be in accordance with MIL-M-26512, including appendix A. The quantitative requirements shall be demonstrated to a confidence level of 80 percent with a 75 percent accuracy level.

4.7 Design data and material

4.7.1 Preproduction test report. The contractor shall prepare a preproduction test report in accordance with MIL-STD-831.

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5. PREPARATION FOR DELIVERY

5.1 The Receiving Set, LORAN AN/ARN-78 shall be prepared for delivery in accordance with the instructions of the procuring activity.

5.2 Reinspection markings. Reinspection marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The Receiving Set, LORAN AN/ARN-78 covered by this specification is a navigation aid intended for use in aircraft and operates at a frequency of 100 kc.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Stock number

c. Preproduction samples and tests

(1) Number of preproduction samples

(2) Point of inspection

(3) Requirement for concurrent delivery of each sample and its test data. The procuring activity should be given at least 10 days prior notice when the preproduction tests are to be conducted so that they may be supervised or witnessed, if desired, by a Government representative.

d. Selection of applicable levels of preservation and packaging, and packing.

6.3 Definitions

6.3.1 Failure. A failure occurs when performance is degraded below that required by this specification. With reference to reliability requirements, minor part failures or faults which do not adversely affect the normal function or performance of the equipment are not to be included as failures; however, such minor failure will be treated as departures from other requirements specified herein or in subsidiary specifications.

6.3.2 LORAN C and LORAN D definitions

6.3.2.1 LORAN C. LORAN C is a long range precision Air Navigation system employing automatic envelope and r-f cycle comparison techniques. It operates in the frequency band 90 to 110 kc with a carrier frequency of 100 kc. It utilizes

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group pulsing (eight pulses/group at each slave station and nine pulses/group at the master station) techniques in lieu of single pulse transmission in order to raise the average transmitted power. The pulses of the group are phase-coded (carrier frequency cycle phase relationship with respect to envelope is changed from pulse to pulse in a group in accordance with a prescribed code). The use of synchronous detection and post-detection narrow banding reduces the effect of noise and interference and in conjunction with the use of phase coding virtually eliminates the effect of skywave contamination. In addition, phase-coding provides signal identification required for automatic search. A LORAN C chain, consisting of one master and two or more slaves, operates on a single specific repetition rate.

6.3.2.1.1 Pulse shape. The LORAN C pulse shape is presently an unsymmetrical pulse having a fast rising (essentially cosine function) leading edge together with a relative slow decay of the trailing edge. It has a total rise time from 0 to 100 amplitude of approximately 70 microseconds. Its total width at 10 percent amplitude is approximately 200 microseconds.

6.3.2.2 LORAN D. LORAN D is a precision Tactical Air Navigation system used for navigation within a 250- to 500-mile area in connection with complimentary ground stations. The system employs automatic envelope and r-f cycle comparison techniques. It operates in the frequency band of 90 to 110 KC with a carrier frequency of 100 KC. It utilizes group pulsing (sixteen pulses/group) techniques in lieu of single pulse transmission in order to raise the average transmitted power. The pulses of the group are phase-coded (carrier frequency cycle phase relationship with respect to envelope is changed from pulse to pulse in accordance with a prescribed code).

6.3.2.3 Basic repetition rate. The basic repetition rate is the approximate rate at which a LORAN transmitter is group pulsed. There are five basic rates, as follows: (Reference 3.6.2b.)

a. SS: 10 pulses per second
b. SL: 12-1/2 pulses per second
c. SH: 16-2/3 pulses per second
d. S: 20 pulses per second
e. L: 25 pulses per second

6.3.2.4 Specific repetition rate. The specific repetition rate is the precise rate at which a LORAN transmitter is group pulsed. Specific rates are identified by a letter(s) followed by a numeral. The letter(s) signify the basic repetition rate, and the numeral indicates the number of hundred of microseconds that the specific period is less than the basic period. Eight specific rates numbered 0 to 7, inclusive, are provided. Repetition

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periods in microseconds for each of the five basic and eight specific rates are tabulated below:

	SS	SL	SH	S	L
0	100,000	80,000	60,000	50,000	40,000
1	99,900	79,900	59,900	49,900	39,900
2	99,800	79,800	59,800	49,800	39,800
3	99,700	79,700	59,700	49,700	39,700
4	99,600	79,600	59,600	49,600	39,600
5	99,500	79,500	59,500	49,500	39,500
6	99,400	79,400	59,400	49,400	39,400
7	99,300	79,300	59,300	49,300	39,300

6.3.2.5 LORAN master station (M). A LORAN master station is one of the group of LORAN transmitting stations comprising a single LORAN C rate. It has the burden of precisely maintaining its assigned repetition rate and carrier frequency, and of monitoring the synchronization, accuracy and stability of its paired slave stations.

6.3.2.6 LORAN slave station (X, Y, Z). A LORAN slave station is the LORAN transmitting station whose transmissions paired with that of its master station comprise a LORAN pair. It has the burden of maintaining an accurate fixed delay between the reception of master and transmission of its own signal, and of monitoring the time difference of the other signals on the same rate.

6.3.2.7 Time difference. Time difference is a measure of the relative times of reception of master and slave signals. It is measured as the time in microseconds between the reception of the master signal and reception of the slave signal.

6.3.2.8 LORAN pulse group

6.3.2.8.1 A LORAN C pulse group consists of 8 (or 9 at a master station) phase-coded pulses transmitted by any one LORAN C station. These 8 pulses are separated by 1000 microseconds within the group. The ninth pulse at a master station is separated from the eight by 2000 microseconds.

6.3.2.8.2 A LORAN D pulse group consists of 16 phase-coded pulses spaced 500 microseconds apart.

6.3.2.9 LORAN group repetition rates (GRR). The LORAN GRR correspond to the specific repetition rates under 6.3.2.4.

6.3.2.10 Phase coding. Phase coding is a method of changing phase of the r-f cycle relative to the envelope of each pulse of the LORAN pulse group.

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The phase relationship varies from pulse to pulse in accordance with the codes specified herein.

6.3.2.10.1 LORAN C phase-coding (in degrees):

Master Pulses

Period	1	2	3	4	5	6	7	8	9
1	0	0	180	180	0	180	0	180	0
2	0	180	180	0	0	0	0	0	180
3	0	0	180	180	0	180	0	180	0
4	0	180	180	0	0	0	0	0	180

Slave Pulses

Period	1	2	3	4	5	6	7	8
1	0	0	0	0	0	180	180	0
2	0	180	0	180	0	0	180	180
3	0	0	0	0	0	180	180	0
4	0	180	0	180	0	0	180	180

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6.3.2.10.2 LORAN D phase-coding (in degrees):

Master Pulses

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0	0	0	0	180	0	180	180	0	0	180	0	0	180	180	0
2	0	0	180	0	180	0	0	180	0	180	0	180	0	0	0	180
3	0	180	0	180	180	180	180	0	0	180	180	180	0	0	180	180
4	0	180	180	180	180	180	0	0	0	0	0	0	0	180	0	0

Slave Pulses

S, Z Slaves

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0	0	0	180	0	0	0	0	0	180	180	0	180	0	0	0
2	0	180	180	0	0	180	180	180	0	180	0	0	180	0	180	0
3	0	180	0	0	0	180	0	180	0	0	180	180	180	180	0	180
4	0	0	180	180	0	0	180	0	0	0	0	180	180	180	180	180

Y Slave

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0	0	0	180	0	0	0	0	0	0	180	180	180	180	0	180
2	0	0	180	180	0	0	180	0	0	180	0	0	180	0	180	0
3	0	180	0	0	0	180	0	180	0	180	180	0	180	0	0	0
4	0	180	180	0	0	180	180	180	0	0	0	180	180	180	180	180

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6.3.2.11 Signal-to-noise ratio. The signal-to-noise (S/N) ratio is defined as the ratio of rms signal at the sampling point to rms noise at the output of the r-f amplifier. The ratio of interfering nonsynchronous cw to LORAN signals is taken at the input to the r-f amplifier (at the sampling point).

6.3.2.12 Signal-to-interference ratio. Signal-to-interference ratio is defined as the square root of the ratio of the power of a pure cw signal having the same peak-to-peak amplitude as that of the r-f cycle of the LORAN pulse at the standard sampling point, to the power of interfering signal.

6.3.2.13 Slave station blink transmission

6.3.2.13.1 LORAN C. All slave stations shall have a common blink procedure. The blink shall consist of the first two of the eight pulses blinking on and off with no shift in position. The duration of the "on" portion shall be between 0.2 and 0.35 second and shall repeat every 4 seconds. The remaining six of the eight pulses shall be continuously "on" with their normal phase code. Blink transmission will indicate a malfunction in system timing accuracy of the transmitter.

6.3.2.13 External velocity information. External velocity information shall be presented to the receiving set over a 4-wire input. Three wires shall provide the velocity information and one wire shall provide the directional information. The information is to be provided within the proproper time sharing interval portion upon proper indication by the LORAN receiver. The pulse scale factors shall represent 49.196 feet/pulse (25.588 feet/pulse in some LORAN D Modes).

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